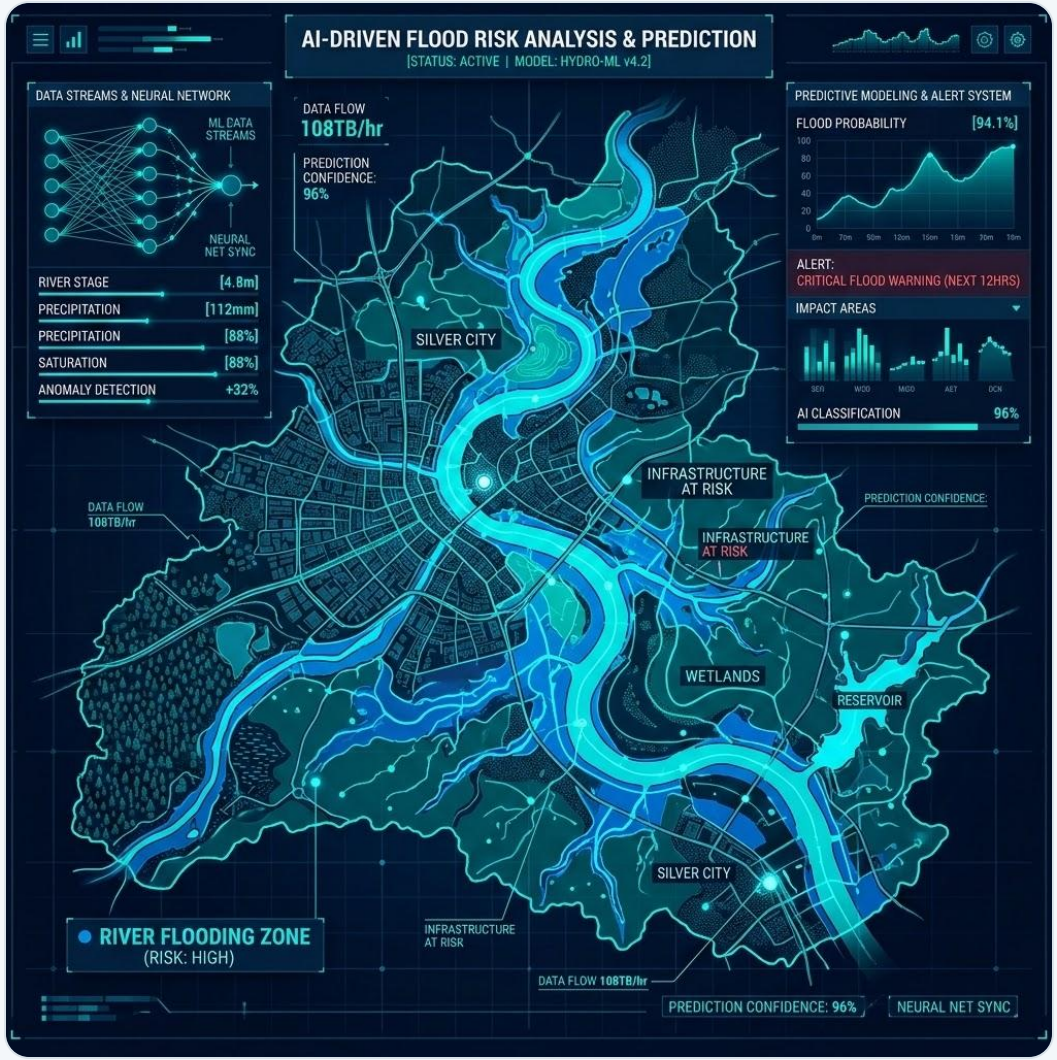
MACHINE LEARNING · CASE STUDY

Predicting Floods with Machine Learning

A real-world case: Google Flood Hub

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ANALYSIS

The problem

01 The Need for Warnings

Floods are one of the most common natural disasters.

To warn people, you need to know the water level before the river rises.

02 The Data Bottleneck

Older methods need many years of data from each single river.

Most rivers in poorer regions have no measuring station at all.

Core Problem: So: no data, no warning — exactly where floods hurt people most.

ANALYSIS

The machine learning solution

Google Flood Hub predicts river floods up to 7 days ahead. It works in over 150 countries and is free to use.

How it learns

It studies rivers that do have measuring stations.

It learns the pattern: this much rain → this much water.

It applies that pattern to rivers with no station.

Two steps

01

Predict how much water will flow in the river.

02

Predict which areas go under water, and how deep.

CONCLUSION

Which type of machine learning?

Selected Approach

Supervised learning

The model predicts the water level. Later, the real water level is measured. That real value is the label, and the model learns from the difference.

Not Applicable

Not unsupervised

We are not looking for hidden groups. We already know what we want to predict.

Not Applicable

Not reinforcement

The model only predicts. It takes no action and gets no reward.

Example: *If the system opened and closed dam gates, that would be reinforcement learning.*

ETHICS

Three questions

01

Who decides when the alarm rings?

Too early and people stop listening. Too late and they are not warned.

The model gives a number — a human chooses the limit. That is a human decision, not a technical one.

02

Not everyone is protected equally

Richer countries have more river data, poorer countries have less.

So the warnings are least reliable where they are needed most.

03

Who is responsible if it fails?

A private company runs a public safety system.

Some scientists say the accuracy is not yet good enough for real warnings, and it is unclear who answers if one fails.

Balance: Before this system, many regions had no warning at all. The open question is who controls it.

SOURCES

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